دائـــــرة الـــصـــحـــة
DEPARTMENT OF HEALTH

# ANTENATAL ULTRASOUND GUIDELINES

July 2022

PUBLIC عـــام

دائـــــرة الـصـــحـــة
DEPARTMENT OF HEALTH

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| **Document Title** | Antenatal Ultrasound Guideline | | |
| **Document Ref. Number** | DOH/Guideline/ Antenatal Ultrasound | **Version** | V.1 |
| **Effective Date** | 14 July 2022 | | |
| **Previous Versions** | N/A | | |
| **Document Owner** | HealthCare Facilities | | |
| **Applies to** | All Healthcare Providers and Professionals licensed by DOH and engaged in the management of maternity patients in the Emirate of Abu Dhabi | | |
| This Standard should be read in conjunction with related UAE laws, DOH standards, policies, and circulars (general and related to Emergency Disaster Management & Preparedness), CBRNE, NCEMA Standards, Policies, DOH Manuals, licensing regulations and reporting requirements. | | | |

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## 1. *Introduction*

*Antenatal ultrasonography has undergone great technological advancement in the last few decades and has become an important diagnostic tool in obstetric practice.*

*Routine ultrasound examination is an established part of modern antenatal care. Apart from routine examinations, ultrasound provides diagnostic findings that often facilitate the management of problems arising during pregnancy, and these clinically indicated scans should be offered when there is a clinical need as guided by the accepted best practice standards.*

*Ultrasound scans should be used as an adjunct to clinical evaluation in pregnancy problems and should not replace clinical examination.*

## 2. *About this Guideline*

*This guideline is based on a review of international guidelines and best practices on antenatal ultrasound. In addition, local technical/expert advice has largely contributed in the development of this guideline in alignment within the local utilization pattern.*

## 3. *Purpose*

3.1. *This guideline is designed to serve as a basic reference for healthcare practitioners performing ultrasound in pregnancy to enhance patient safety and facilitate a high-quality ultrasound service for pregnant women.*

3.2. *Judicious use of ultrasound scans is important to reduce unnecessary anxiety and intervention. All non-routine scans shall be justified clinically and indication documented in the patient record. Refer to Appendix 1 for a list of ultrasound scans that can be ordered, their description and associated CPT code.*

## 4. *Scope*

*4.1 This guideline can be used by all healthcare providers and professionals licensed by DOH and engaged in the management of maternity patients in the Emirate of Abu Dhabi. It relates to all sonographers, trained obstetricians as well as radiologists who perform ultrasound examinations on pregnant women.*

*4.2 This guideline applies to all women undergoing ultrasound examination at any stage of pregnancy or who are receiving follow up care after delivery or pregnancy loss.*

*4.3 The guidance covers*

*4.3.1 Recommended criteria for performing each type of scan*

*4.3.2 Advanced obstetric ultrasound usually undertaken under the auspices of fetal medicine services.*

## 5. *Abbreviations and Definitions*

| *Category* | *Definition* |
|---|---|
| *Fetal Growth Restriction (FGR)* | *Fetal growth restriction (FGR) is a condition in which an unborn baby (fetus) is smaller than expected for the number of weeks of pregnancy (gestational age). It's often described as an estimated weight less than the 10th percentile.* |
| *Nuchal Translucency (NT)* | *Nuchal translucency (NT) is the sonographic appearance of a collection of fluid under the skin behind the fetal neck in the first-trimester of pregnancy.* |
| *Retained Products of Conception (RPOC)* | *Retained products of conception (RPOC) refers to fetal or placental tissue that remains in the uterus after a pregnancy.* |
| *Small Gestational Age (SGA)* | *Small-for-gestational age (SGA) refers to an infant born with a birth weight less than the 10th centile.* |

# 6. Criteria for Different Ultrasound Examinations in Pregnancy

## 6.1. *Routine Ultrasound Scan in Pregnancy*

*6.1.1. All pregnant women are offered three routine scans in pregnancy as part of the antenatal care. The aim of these scans is to confirm viability, normal location of pregnancy, appropriate dating, to exclude major malformations and to ensure normal fetal growth and wellbeing. However, there are limitations with these examinations and to gain the maximum benefit, these scans should be appropriately timed.*

*6.1.2. The first scan should be performed in 1st trimester ideally at 11-13 weeks and 6 days to accurately date the pregnancy (refer to Appendix 2 for gestational dating guidance), to confirm number of fetuses and to exclude major malformations. If indicated, NT measurement can be offered at the same time as screening test for chromosomal abnormalities.*

*6.1.3. The second scan should be offered in mid trimester at 18 - 22 weeks for detailed anatomy assessment.*

*6.1.4. The third routine scan should be offered in third trimester to exclude small for gestational age (SGA) and fetal growth restriction (FGR). The Route study concluded that in low-risk pregnancies, routine ultrasound examination at 36 weeks' gestation was more effective than that at 32 weeks' gestation in detecting FGR and related adverse perinatal and neonatal outcomes. Hence the routine third trimester scan should be offered at 34-36 weeks for best outcome.*

## 6.2. *First Trimester Ultrasound Examination*

*6.2.1 For pregnancy < 8 weeks gestation, transvaginal scan (TVS) is advised to evaluate the pregnancy and other pathologies accurately. Transvaginal scan should also be advised in pregnancy >8 weeks if transabdominal evaluation is suboptimal due to limiting factors such as maternal obesity, uterine position, scarred abdomen etc.*

*6.2.2 Scan objectives:*

- *Identify and assess the viability*
- *Establish the number and location of intrauterine gestations and assess chorionicity and amnionicity in multiple pregnancy*
- *Establish gestational age (refer to Appendix 2);*
- *Assess maternal pelvis for any evidence of extra-uterine pregnancy;*
- *Detect gross fetal abnormalities;*
- *Measure the nuchal translucency (NT) if appropriate;*
- *Assess for retained products of conception (RPOC) if appropriate;*
- *Assess the adnexa for any pathology;*

### 6.3. *Second Trimester Ultrasound*

*6.3*

*6.3.1 The main objective of a routine mid-trimester fetal ultrasound scan is to provide accurate diagnostic information for the delivery of optimized antenatal care with the best possible outcomes for mother and fetus. The procedure is used to primarily detect structural fetal abnormalities. Additionally, early fetal growth is evaluated and if not yet determined gestational dating undertaken (see Appendix 2).*

*6.3.2 Non-routine second trimester scan: A medical indication may arise whereby a scan is required as part of clinical evaluation before the routine mid-trimester scan e.g. vaginal bleeding, abdominal or pelvic pain. This scan would include an assessment of fetal biometry, liquor volume, placental site, maternal adnexa and evaluation for evidence of bleeding but would not replace the need for anatomy scan as described in 6.4.*

### 6.4. *Detailed anatomy scan*

*6.4*

*6.4.1 A routine mid-trimester ultrasound scan is ideally performed between 18 and 22 weeks of gestation. This period represents an ideal gestation to detect major fetal abnormalities.*

*6.4.2 Recommended minimum requirements for a basic fetal anatomical survey during the mid-trimester of pregnancy are summarized in Appendix 3.*

*6.4.3 If the scan is not complete, a repeat limited scan should be undertaken in 2-3 weeks to attempt completion.*

### 6.5. *Third trimester Ultrasound*

*6.5*

*6.5.1 Ultrasound examination in third trimester mainly focuses on evaluation of fetal wellbeing i.e. fetal growth, liquor volume and umbilical artery Doppler in some cases (Appendix 4).*

*6.5.2 In low risk pregnancy one routine scan is undertaken in third trimester, ideally at 34-36 weeks gestation.*

*6.5.3 Women at high risk of fetal growth pathology (growth restriction or macrosomia) should have serial scans to evaluate the growth velocity and early detection of deviation from the normal growth curve. (Refer to Appendix 5 for a list of indications for serial growth scans in high risk pregnancy).*

*6.5.4 Placental localization in relation to the internal cervical os should be described in all second and third trimester scans.*

*6.5.4.1 Transvaginal scan (TVS) is superior to transabdominal scan (TAS) to diagnose placenta previa and low-lying placenta, particularly in the third trimester.*

*6.5.4.2 The term placenta Previa should be used if the placenta lies directly over the internal os.*

*6.5.4.3 If the placenta is thought to be low lying in the routine anatomy scan, a follow up abdominal scan should be arranged at 28-30 weeks to assess if the placenta is still low. If it is unclear on TAS, TVS should be performed.*

*6.5.4.4 If the placenta is still low at 28-30 weeks (covering, reaching or < 20mm from internal os), a TV scan should be arranged at 4 weeks later to confirm placental location (TVS is mandatory). In addition to TVS a TAS should be undertaken to assess fetal wellbeing and extent of placenta.*

*6.5.4.5 In women with a persistent low-lying placenta or placenta previa at 32 weeks of gestation who remain asymptomatic, an additional fetal growth scan (TAS) and TVS is recommended at around 36 weeks of gestation to inform discussion about mode of delivery.*

*6.5.4.6 No definitive diagnosis of low-lying or placenta previa should be made from 32 weeks onward without transvaginal scanning.*

*6.5.4.7 For women with previous CS or uterine surgery, the 28, 32- and 36-weeks scans should be undertaken in fetal medicine services to enable evaluation for abnormally invasive placenta and management thereof.*

*6.5.4.8 In women presenting with antepartum hemorrhage, TVS should be performed in addition to transabdominal scan to confirm/exclude placenta previa and low-lying placenta.*

*6.5.4.9 Where there is, an amniotic fluid abnormality, there should be referral to fetal medicine services for further evaluation.*

6.6. *Timing of examination for late bookers*

*6.6*

*6.6.1 If the women had a good quality anatomy scan in another hospital, this should not be repeated unless there are concerns.*

*6.6.2 If the woman has not had a good quality anomaly scan or has not had a scan at all, an attempt should be made to evaluate fetal anatomy but it should be acknowledged that complete evaluation may not be possible and no further attempt advised*

6.7. *Interval between examinations*

*6.7*

*6.7.1 Repeat biometric measurements should not be taken less than 14 days from previous scan.*

*6.7.2 Abnormal liquor volume should be reassessed 1-2 weekly depending on clinical circumstances unless it reverts to normal.*

*6.7.3 Abnormal umbilical artery Doppler should be re-evaluated at least weekly for confirmed SGA/FGR Cases (but this is usually in the fetal medicine service).*

6.8. *Cervical length assessment*

*6.8*

*6.8.1 Preterm birth causes significant neonatal morbidity and mortality, as well as long-term disability. Therefore, strategies for preventing preterm birth are important.*

*6.8.2 Transvaginal cervical length measurement is a useful tool in the prediction of preterm labour in women at risk. It should be performed in certain high-risk cases at least fortnightly from 14-26 weeks (see Appendix 4).*

6.9. *Ultrasound in Multiple Pregnancy*

*6.9*

*6.9.1 In multiple pregnancy there is a high risk of fetal growth pathology and routine clinical examination cannot be used to screen for this.*

*6.9.2 Furthermore, if there is monochorionicity there is a risk of complications of shared placentation (fetal-to-fetal transfusion syndrome, selective growth restriction, twin anemia polycythemia syndrome and twin reversed arterial perfusion sequence).*

*6.9.3 Therefore, in addition to routine first trimester and second trimester anatomy scan, serial ultrasonography is required and the frequency and elements of the scan depend on the type of multiple pregnancy. Refer to Appendix 4 for guidance about frequency of ultrasound in multiple pregnancy.*

## *7. Responsibilities*

*7.1 Where ultrasound findings are not normal it is the responsibility of the ordering physician to evaluate the clinical situation and plan management which may include referral to fetal medicine services.*

*7.2 In some cases, the initial / routine imaging should be undertaken in the fetal medicine setting. See Appendix 6 for indications for referral to fetal medicine services.*

## 8. Reviewers:

| Name of Reviewer | Profession | Organization |
|---|---|---|
| Dr Leanne Bricker | Chair, Department of Fetal Medicine & Imaging | Corniche SEHA |
| Dr. Ali Al Ibrahim | Consultant, Maternal Fetal Medicine | Kanad Hospital |
| Dr. Abdulrauf Ghulam | Supervisor, Medical Standards and Research | Daman |
| Dr. Dina Ahmed Hewaidy | Advisor, Clinical Review and Investigations | Department of Health Abu Dhabi |

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9.6. *Roma E, Arnau A, Berdala R, Bergos C, Montesinos J, Figueras F. Ultrasound screening for fetal growth restriction at 36 vs 32 weeks' gestation: a randomized trial (ROUTE). Ultrasound Obstet Gynecol. 2015; 46(4):391-397. doi:10.1002/uog.14915.*

9.7. *RCOG Green Top Guideline no 27a, Sept 2018, Placenta praevia and placenta praevia accrete: Diagnosis and Management. Available at https://www.rcog.org.uk/en/guidelines-research-services/guidelines/gtg27a/.*

9.8. *RCOG Green Top Guideline no 31, revised Jan 2014, The investigation and management of the Small-for-Gestational-Age Fetus. https://www.rcog.org.uk/en/guidelines-research-services/guidelines/gtg31/.*

9.9. *UK NICE Clinical Guideline (NG137): Twin and Triplet pregnancy. NICE guideline Published: 4 September 2019 www.nice.org.uk/guidance/ng137.*

9.10. *RCOG Green Top Guideline number 51. Kilby M, Bricker L. Management of Monochorionic Twin Pregnancy. BJOG 2016; 124: e1-e45*

9.11. *UK NICE Clinical Guideline (NG25): Preterm labour and birth, published 2015, updated 2019 https://www.nice.org.uk/guidance/ng25.*

9.12. *Berghella V. Second-trimester evaluation of cervical length for prediction of spontaneous preterm birth in singleton gestations. In: UpToDate, Lockwood CJ, Levine D (Ed), UpToDate, Waltham, MA, 2014. Updated May 1, 2019. Accessed May 17, 2019.*

## 10. **Appendices**

| |
|---|
| 1. *Ultrasound in Pregnancy List with CPT Codes and CPT Code Descriptions* |
| 2. *Gestational Dating Guideline* |
| 3. *Guidance on Elements of Anatomy Scan Including Images to Be Stored* |
| 4. *Guidance on the Indications for US Follow up and Doppler* |
| 5. *Guidance on the Indications for Serial Growth Scan (i.e. Medical Necessity Scan) in High Risk Pregnancy* |
| 6. *Referral Criteria for Fetal Medicine Services* |

**Appendix 1: Ultrasound in Pregnancy List with CPT Codes, CPT Code Descriptions and Clinical Description:**

| CPT | CPT Code Description | Clinical Description |
|---|---|---|
| **76801** | *Ultrasound, pregnant uterus, real time with image documentation, fetal and maternal evaluation, first trimester (< 14 weeks 0 days), transabdominal approach; single or first gestation* | *First trimester scan* |
| **76802** | *Ultrasound, pregnant uterus, real time with image documentation, fetal and maternal evaluation, first trimester (< 14 weeks 0 days), transabdominal approach; each additional gestation (List separately in addition to code for primary procedure)* | *First trimester scan, for additional fetuses in multiple pregnancy* |
| **76805** | *Ultrasound, pregnant uterus, real time with image documentation, fetal and maternal evaluation, after first trimester (> or = 14 weeks 0 days), transabdominal approach; single or first gestation* | *Routine growth scan or late booking scan* |
| **76810** | *Ultrasound, pregnant uterus, real time with image documentation, fetal and maternal evaluation, after first trimester (> or = 14 weeks 0 days), transabdominal approach; each additional gestation (List separately in addition to code for primary procedure)* | *Routine growth scan or late booking scan, for additional fetus in multiple pregnancy* |
| **76811** | *Ultrasound, pregnant uterus, real time with image documentation, fetal and maternal evaluation plus detailed fetal anatomic examination, transabdominal approach; single or first gestation* | *Routine anatomy scan* |
| **76812** | *Ultrasound, pregnant uterus, real time with image documentation, fetal and maternal evaluation plus detailed fetal anatomic examination, transabdominal approach; each additional gestation (List separately in addition to code for primary procedure)* | *Routine anatomy scan, for additional fetuses in multiple pregnancy* |
| **76813** | *Ultrasound, pregnant uterus, real time with image documentation, first trimester fetal* | *NT scan* |

| | | |
|---|---|---|
| | *nuchal translucency measurement, transabdominal or transvaginal approach; single or first gestation* | |
| **76814** | *Ultrasound, pregnant uterus, real time with image documentation, first trimester fetal nuchal translucency measurement, transabdominal or transvaginal approach; each additional gestation (List separately in addition to code for primary procedure)* | *NT scan , for additional fetus in multiple pregnancy* |
| **76815** | *Ultrasound, pregnant uterus, real time with image documentation, limited (eg, fetal heartbeat, placental location, fetal position and/or qualitative amniotic fluid volume), 1 or more foetuses* | *Quick look scan* |
| **76816** | *Ultrasound, pregnant uterus, real time with image documentation, follow-up (eg, re-evaluation of fetal size by measuring standard growth parameters and amniotic fluid volume, re-evaluation of organ system(s) suspected or confirmed to be abnormal on a previous scan), transabdominal approach, per fetus* | *Follow up scan if fetal anomaly or growth pathology or high risk of growth pathology* |
| **76817** | *Ultrasound, pregnant uterus, real time with image documentation, transvaginal* | *For early pregnancy, cervical length or placental site assessment* |
| **76818** | *Fetal biophysical profile; with non-stress testing* | *Biophysical profile with CTG* |
| **76819** | *Fetal biophysical profile; without non-stress testing* | *Biophysical profile without CTG* |
| **76820** | *Doppler velocimetry, fetal; umbilical artery* | *For small or growth restricted fetuses* |
| **76821** | *Doppler velocimetry, fetal; middle cerebral artery* | *For small or growth restricted fetuses or assessment for suspected or confirmed fetal anaemia* |
| **76825** | *Echocardiography, fetal, cardiovascular system, real time with image documentation (2D), with or without M-mode recording;* | *This represents the grayscale / 2D element* |
| **76826** | *Echocardiography, fetal, cardiovascular system, real time with image documentation (2D), with or without M-mode recording; follow-up or repeat study* | *This represents the Colour flow Doppler* |
| **76827** | *Doppler echocardiography, fetal, pulsed wave* | *This represents the grayscale* |

| | | |
|---|---|---|
| | *and/or continuous wave with spectral display; complete* | */ 2D element* |
| **76828** | *Doppler echocardiography, fetal, pulsed wave and/or continuous wave with spectral display; follow-up or repeat study* | *This represents the Colour flow Doppler* |
| **93325** | *Doppler echocardiography colour flow velocity mapping (List separately in addition to codes for echocardiography)* | *This represents the pulsed Doppler* |

**Appendix 2**: **Gestational Dating Guideline**

1. *First trimester*
   - *1.1 If patient is certain of the date of her last menstrual period (LMP) and has a regular menstrual cycle, every 26 - 33 days, use the expected date of delivery (EDD) calculated based on the LMP unless there is variation as per point 1.3.*
   - *1.2 If the patient is unsure of her LMP, dating should be based on ultrasound estimates using crown-rump length measurement (CRL), ideally before or at 13 weeks and 6 days of gestation, with the earliest ultrasound with a crown-rump length (CRL) measurement being the most reliable. If the crown-rump length is above 84 mm, the gestational age should be estimated using head circumference measurement (HC).*
   - *1.3 If there is variation of > 5 days in the calculation between the LMP and first trimester ultrasound, the EDD should be adjusted based on first trimester ultrasound dating.*

2. *Second trimester*
   - *2.1 The crown-rump length (CRL) measurement for estimation of gestational age becomes less accurate when the CRL is > 84 mm (corresponding to approximately 14 weeks of gestation), and in these cases head circumference (HC), should be used to estimate the EDD.*
   - *2.2 Dating changes for smaller discrepancies are appropriate based on how early in the first trimester the ultrasound examination was performed and clinical assessment of the reliability of the LMP date.*
   - *2.3 If dating by ultrasonography performed between 14 weeks and 16 weeks of gestation, inclusive, varies from LMP dating by more than 7 days, or if ultrasound dating between 17 weeks and 22 weeks of gestation varies by more than 10 days, the EDD should be changed to correspond with the ultrasound dating. (If there is no first trimester dating ultrasound).*
   - *2.4 Between 22 weeks and 28 weeks of gestation, ultrasound dating has an accuracy of ±10 – 14 days. If ultrasound dating between 22 weeks and 28 weeks of gestation, inclusive, varies by more than 14 days from LMP dating, the EDD should be changed to correspond with the ultrasound dating (if there is no first trimester dating ultrasound).*

3. *Third trimester*
   - *3.1 Gestational age assessment by ultrasonography in the third trimester (28 weeks of gestation and beyond) is the least reliable method, with an accuracy of ± 21 – 30 days.*
   - *3.2 The ultrasound parameter to use in the third trimester to estimate EDD is HC. Femur length (FL) or trancerebellar diameter (TCD) may also be used.*
   - *3.3 Because of the risk of re-dating a small for gestational age fetus that may be growth restricted, management decisions based on third-trimester ultrasonography alone are especially problematic; they need to be guided by careful consideration of the entire clinical picture and may require closer surveillance, including repeat ultrasonography to ensure appropriate interval growth.*
   - *3.4 The best available data to support adjusting the EDD of a pregnancy if the first ultrasonography in the pregnancy is performed in the third trimester and suggests a discrepancy in gestational dating of more than 21 days.*

*3.5 If pregnancy is dated in the third trimester, sonography should be repeated 2 - 3 weeks later to ensure appropriate interval growth.*

4. *Special Circumstances*

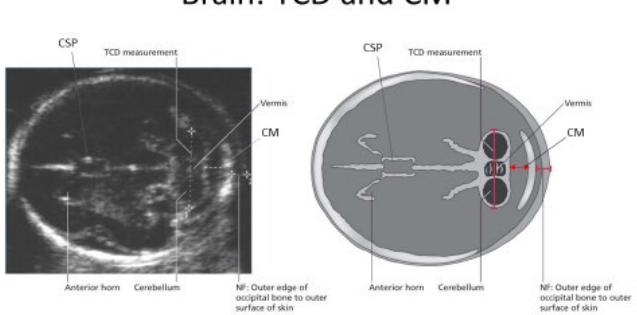
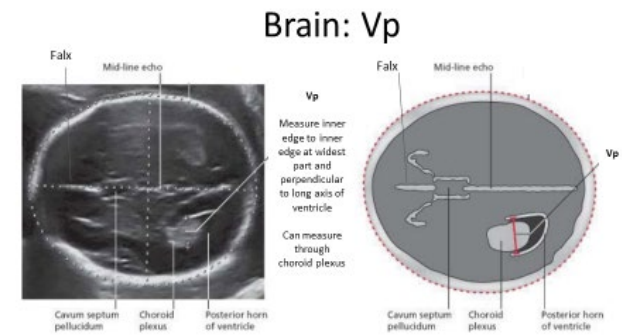
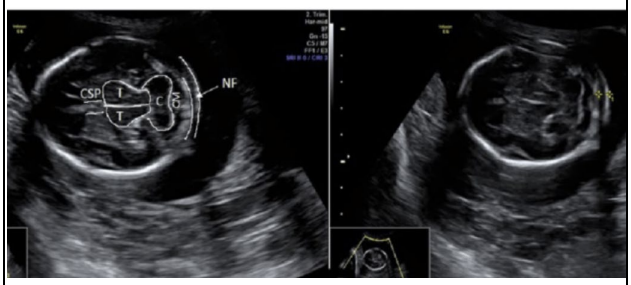
*4.1 If the pregnancy resulted from assisted reproductive technology (ART), the ART derived gestational age should be used to assign EDD.*

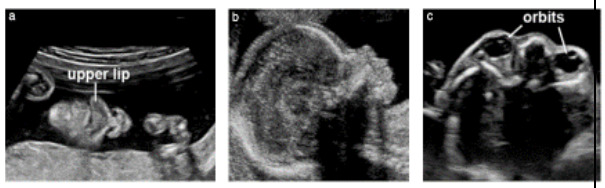
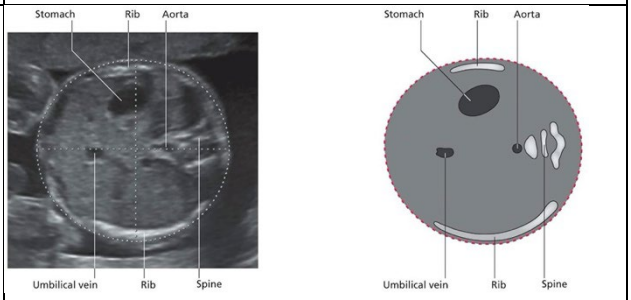
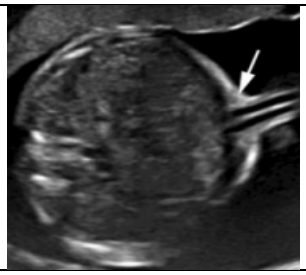
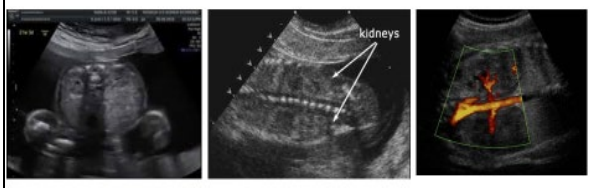
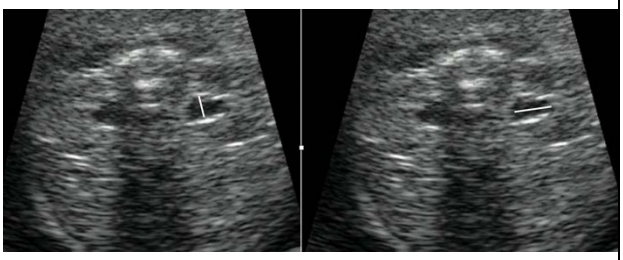
*4.2 The ART derived gestational age is based on date of oocyte retrieval i.e. age of the embryo. EDD is 266 days from oocyte retrieval.*

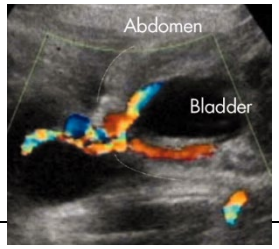
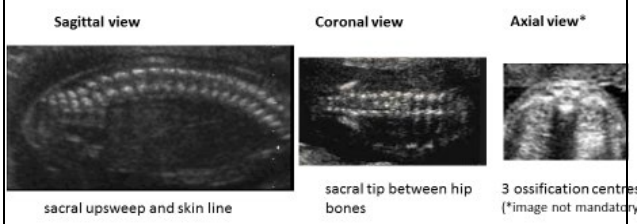
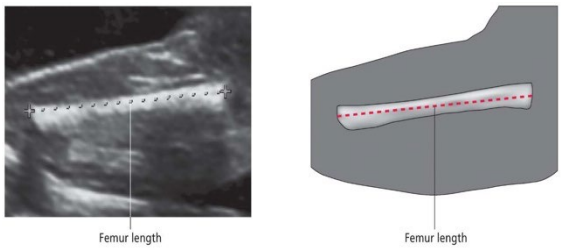
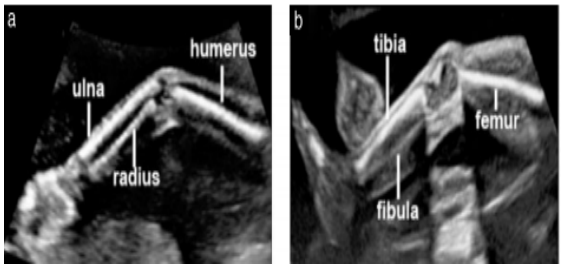
*4.3 A practical method to calculate EDD is to minus 14 days from oocyte retrieval and assume it is 'LMP' e.g. oocyte retrieval was 15 September, use dating 'LMP' from LMP as 1 September.*

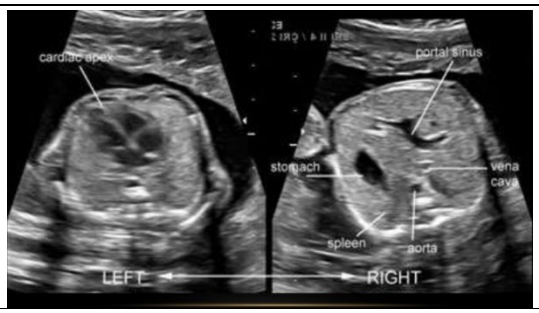
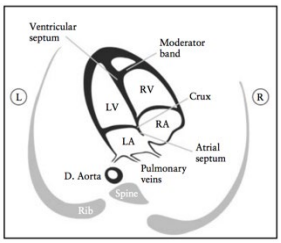
*4.4 In multiple pregnancy use the CRL or HC (depending on trimester as in 2.1, 2.2 and 2.3) of the largest fetus.*

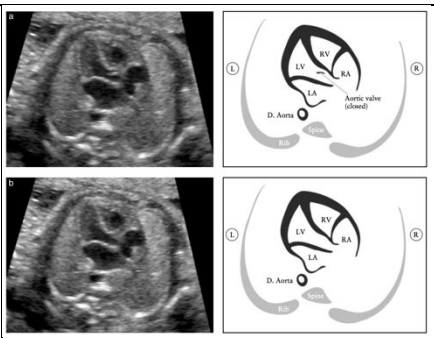
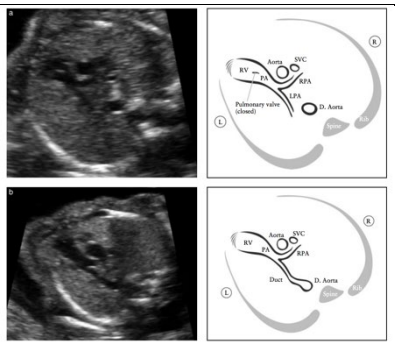
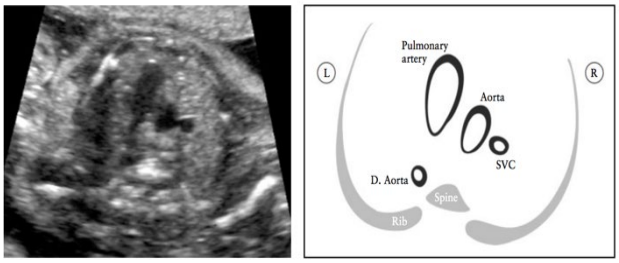
**Appendix 3: Guidance on Elements of Anatomy Scan including Images to be Stored** (ISUOG Practice Guidelines)

| *Structure* | *Section* | *Caliper placement* | *Images capture & archive* |
|---|---|---|---|
| *Head circumference (HC)* | *Transthalamic plane through CSP* | *Derived ellipse around outer border* |  |
| *Transverse cerebellar diameter (TCD)* | *Transcerebellar plane through CSP* | *Linear, maximum diameter, outer to outer border* |   |
| *Atrium (Vp)* | *Transventricular plane, level of the glomus of the choroid plexus through CSP* | *Inner border of medial wall to inner border of outer wall, perpendicular to ventricular cavity of distal ventricle (measure the widest area even if through posterior portion of choroid plexus)* |   |
| *Nuchal skin fold* (NF)* | *Transcerebellar plane* | *Outer border of occiput to outer border of overlying skin* |   |

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| *Face* | *Coronal view: lips and nasal tip*<br>*Transverse/coronal view at the level of the orbits*<br>*Midsagittal profile view** | *Measurements not required* |  <br>Mandatory image *Profile – not mandatory but please do if possible Mandatory image |
| *Abdominal circumference (AC)* | *Stomach and short intrahepatic section of umbilical vein.* | *Derived ellipse* |  |
| *Abdominal wall* | *Insertion of umbilical cord into fetal abdomen.* | *Measurements not required* |  |
| *Right and Left Kidney* | *Transverse section of each, demonstrating maximum diameter of AP renal pelvis*<br>*Optional coronal/sagittal views.* | *If suspected dilatation (defined as >/=6mm) measure AP pelvis - inner border of anterior wall to inner border of posterior wall of renal pelvis* |  <br>Tips:<br>- If difficult to see - change colour eg. use sepia<br>- If difficult to see - check for renal blood flow in coronal view<br>  <br>Correct measurement Incorrect measurement |

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| *Bladder and 3 vessel cord* | *Cross section demonstrating maximum size (must be seen).* | *Measurements not required*<br>*Use colour Doppler to demonstrate 2 umbilical arteries* |  |
| *Spine* | *Sagittal section including sacrum and demonstrating skin covering.*<br>*Coronal views of the full length of the fetal spine and demonstrating that sacrum sits between hip bones.*<br>*Axial views from top to bottom (but image not mandatory).* | *Measurements not required.* |  |
| *Femur (FL)* | *Full femoral displays at 90o to ultrasound beam.* | *Linear measurement from one end of the diaphysis to other, disregarding the curvature.* |  |
| *Extremities* | *Presence of metacarpals & metatarsals*<br>*Assess the presence of all 3 long bones & their alignment in 2 arms and 2 legs.* | *Measurements not required*<br>*But measure if they appear short or abnormal.* |  |

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| *Diaphragm* | *Sagittal or coronal view demonstrating the thorax & abdomen with echo poor concave line of diaphragm separating both (with stomach below diaphragm).* | *Not required.* |  |
| *Placental site* | *Midline sagittal section of uterus demonstrating relationship between leading edge of placenta and (assumed) internal os – label image as demonstrated.* | *Measurements not required.* |  |
| *Fetal heart: situs* | *Demonstrate situs i.e. heart and stomach on left. Preferably double image or alternatively one image after the other.* | |  |
| *Four chamber view: transverse section of the thorax including a complete rib and crux of the heart.* | *Normal cardiac size occupies 1/3rd chest.*<br>*2 atria and ventricles of equal size. 2 patent atrioventricular valves of equal size.*<br>*Differential offsetting of valves, the tricuspid inserts more apically* | *Mostly in the left chest*<br>*Apex points towards the left* |   |

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| *Fetal heart: left ventricular outflow tract* | *Aorta arises from the left ventricle and sweeps outward towards the right shoulder.* | *The anterior wall of the aorta is continuous with the ventricular septum.* |  |
| *Fetal heart: right ventricular outflow tract* | *Main pulmonary artery arises from the right ventricle directed backwards towards the spine. The pulmonary artery lies to the left with the superior vena cava to the right and the aorta in the middle.* | *The main pulmonary artery bifurcates.* |  |
| *Fetal heart: 3 vessel view* | *The diameter of the pulmonary artery is slightly greater than that of the aorta which is slightly greater than that of the superior vena* | |   |
| *3 vessel and trachea view** | *Pulmonary artery continues as arterial duct and aortic arch to the left of trachea continuing as descending aorta; ductus joining the* | *Optional additional view recommended* |   |

*Note: Items marked with * are desirable, not mandatory.*

**Appendix 4: Medical indications for additional ultrasound in pregnancy including Doppler**

| *Indications for US Follow up and Doppler* | |
|---|---|
| *Cervical length assessment for Preterm Birth Screening* | • *One of the Risk Factors below:*<br>- *History of Previous Preterm Birth.*<br>- *Previous Cervical Surgery.*<br>- *Congenital uterine Abnormalities- e.g. Bicornuate Uterus with Pregnancy risk of Preterm Birth.*<br>- *Short Cervix.*<br>- *Mid trimester Pregnancy loss.*<br>• *Trans Vaginal Ultrasound 76817 cervical length to be done from 14 weeks. Gestation until 26 weeks gestation.*<br>• *Cervical length ≤30mm, can repeat US weekly.*<br>• *Cervical length ≥30mm, can Repeat US every 2-4 weeks.* |
| *SGA Screening and FGR screening (risk factors for small babies)* | • *Follow Up Ultrasound CPT 76816, from 24-28 weeks of gestation every 2-4 weeks to determine the rate of growth.*<br><br>• *CPT 93975 or 93976 (Uterine Artery Doppler), from 16-24 weeks of gestation can be used Only once if necessary to assess uteroplacental function. Repeating Uterine Artery Doppler is not Recommended.*<br><br>• *Risk factors:*<br>- *Pre-existing conditions:*<br>▪ *Pre-existing diabetes*<br>▪ *Maternal hypertension*<br>▪ *Maternal autoimmune disease e.g. SLE, antiphospholipid syndrome, Rheumatoid arthritis*<br>▪ *Maternal renal disease including after transplant*<br>▪ *Maternal heart disease (congenital or acquired)*<br>▪ *Maternal age >/= 40*<br>▪ *Maternal smoker >10 per day*<br>▪ *Maternal daily vigorous exercise*<br>▪ *Previous bariatric surgery*<br>- *Previous Obstetric history:*<br>▪ *Previous stillbirth*<br>▪ *Previous fetal growth pathology (SGA or IUGR)*<br>- *Obstetric complications:*<br>▪ *Gestational diabetes*<br>▪ *Gestational hypertension requiring antihypertensive medication*<br>▪ *Pre-eclampsia*<br>▪ *Obstetric cholestasis*<br>▪ *Recurrent antepartum haemorrhage (>1 haemorrhage in second* |

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| | *or third trimester)*<br>▪ *COVID19 requiring hospitalization in index pregnancy*<br>- *Fetoplacental abnormalities:*<br>▪ *Fetal echogenic bowel*<br>▪ *1 or more major fetal structural abnormality e.g. congenital heart disease, CNS malformation*<br>▪ *Placenta praevia (major)*<br>▪ *Placental or cord tumour or cyst*<br>- *Test results:*<br>▪ *PAPP-A < 0.40 MoM*<br>▪ *AFP > 2.0 MoM in second trimester*<br>▪ *Abnormal uterine artery Doppler (defined as mean PI >/= 2.0 in first trimester or >/= 1.45 in second trimester or bilateral notching at any gestation.*<br>- *Clinically Examination Small for gestational dates (clinically suspected IUGR)*<br>- *Unsuitable for monitoring fetal growth by clinical abdominal examination:*<br>▪ *Maternal BMI >/= 35*<br>▪ *Multiple fibroids or 1 large fibroid (>8cm in size)*<br>- *3 or more of the following minor risk factors:*<br>▪ *Maternal age >35yrs*<br>▪ *IVF Singleton Pregnancy*<br>▪ *Nulliparity*<br>▪ *BMI <20*<br>▪ *BMI 25-34.9*<br>▪ *Smoker 1-10/day*<br>▪ *Previous Pre-eclampsia*<br>▪ *Pregnancy Interval <6 months*<br>▪ *Pregnancy Interval >60 months* |
| *Confirmed SGA and FGR* | • *AC or EFW 3rd -10th percentile with normal Umbilical Artery and Middle Cerebral Artery Doppler (SGA):*<br>- *Follow Up Ultrasound CPT 76816, from diagnosis every 2 weeks to determine the rate of growth.*<br>- *CPT 76820 (Umbilical Artery Doppler Ultrasound) and CPT 76821 (MCA Doppler) from diagnosis every 2 weeks. From the time of Diagnosis until Delivery and all are done at the same time.*<br>- *CPT 76818/76819 – weekly.*<br><br>• *EFW / AC < 3rd percentile OR EFW / AC 3rd - 10th percentile with abnormal Umbilical Artery Doppler and / or abnormal Middle Cerebral Artery Doppler (FGR):* |

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| | - *CPT 76816,76820,76821,76818 or 76819 follow up weekly.*<br>- *CPT 93975 or 93976 (Uterine Artery Doppler) Only once at any stage of gestation.* |
| *Amniotic fluid abnormalities (Polyhydramnios, Oligohydramnios, PPROM (preterm prelabour rupture of membranes)* | • *Follow up Ultrasound CPT 76816, every 2-4 weeks, from 2nd trimester onwards.* |
| *Red cell antibody in Pregnancy (Risk of HDFN)* | • *Red blood cell antibodies identified and (or) History of pregnancy complicated by RBC Alloimmunization.*<br>• *Previous OB US report/Scan indicating Fetal anaemia (e.g. ascites, pleural effusions, hydrops, placentomegaly).*<br>• *Follow Up Ultrasound CPT 76816 and CPT 76821 (MCA Doppler). Repeat Every 1-2 weeks from 16 weeks until delivery.* |
| *Large for Gestational Age* | • *EFW >90th Percentile:*<br>- *Follow Up Ultrasound CPT 76816, every 2-4 weeks from 3rd trimester until delivery.* |
| *Multiple Gestations* | • *Dichorionic Diamniotic Twin Pregnancy:*<br>- *Follow up OB US CPT 76816, from 24-36 weeks of gestation, every 4 weeks.*<br>- *Umbilical artery CPT 76820: From 20 weeks of gestation, every 4 weeks until delivery.*<br>- *Cervical length assessment CPT 76817 once between 16-24 weeks gestation.*<br>• *Monochorionic Diamniotic Twin Pregnancy:*<br>- *Follow up OB US CPT 76816, from 16weeks of gestation, every 2 weeks until delivery.*<br>- *Umbilical Doppler CPT 76820 and MCA Doppler CPT 76821: From 20 weeks of gestation, every 2 weeks until delivery.*<br>- *Cervical length assessment CPT 76817 once between 16-24 weeks gestation.*<br>• *Trichorionic Triamniotic Triplet Pregnancy:*<br>- *Follow up OB US CPT 76816 from 16 weeks of gestation, every 2 weeks* |

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| | *until delivery.*<br>- *Umbilical artery CPT 76820: From 20 weeks of gestation, every 2 weeks until delivery.*<br>- *Cervical length assessment CPT 76817 at 16-17, 20-21 and 22-24 weeks gestation.*<br><br>• *Dichorionic Triamniotic or Monochorionic Triamniotic Triplet pregnancy:*<br>- *Follow up OB US CPT 76816 from 16 weeks of gestation until delivery.*<br>- *Umbilical Doppler CPT 76820 and MCA Doppler CPT 76821: From 20 weeks of gestation, every 2 weeks until delivery.*<br>- *Cervical length assessment CPT 76817 at 16-17, 20-21- and 22-24-weeks' gestation.*<br>• *Quadruplets or more (same as for triplet pregnancy).* |
| *Placental Abnormalities* | • *Document on Placenta Previa or Placenta Accreta or Vasa Previa.*<br><br>• *Follow up OB US CPT 76816 and/or TVS CPT 76817 at 28-36 weeks of gestation.* |

**Appendix 5: Guidance on the Indications for Serial Growth Scan (i.e. Medical Necessity Scan) in High Risk Pregnancy**

*The following are risk factors for fetal growth pathology and serial fetal growth scans are medically indicated every 2-4 weeks from 24-26 weeks gestation. It might be more appropriate to do the growth scan every 2 weeks after 32-34 weeks.*

- *Pre-existing conditions:*
  - *Pre-existing diabetes*
  - *Maternal hypertension*
  - *Maternal autoimmune disease e.g. systemic lupus erythematosus (SLE), antiphospholipid syndrome, rheumatoid arthritis*
  - *Maternal renal disease including after transplant*
  - *Maternal heart disease (congenital or acquired)*
  - *Maternal age >/= 40*
  - *Maternal smoker >10 per day*
  - *Maternal daily vigorous exercise*
  - *Previous bariatric surgery*

  *Previous Obstetric history:*
  - *Previous stillbirth*
  - *Previous fetal growth pathology (SGA or IUGR)*

  *Obstetric complications:*
  - *Gestational diabetes*
  - *Gestational hypertension requiring antihypertensive medication*
  - *Pre-eclampsia*
  - *Obstetric cholestasis*
  - *Recurrent antepartum haemorrhage (>1 haemorrhage in second or third trimester)*
  - *COVID19 requiring hospitalization in index pregnancy*

- *Fetoplacental abnormalities*
  - *Fetal echogenic bowel*
  - *1 or more major fetal structural abnormality e.g. congenital heart disease, CNS malformation*
  - *Placenta praevia (major)*
  - *Placental or cord tumour or cyst*

- *Test results:*
  - *PAPP-A < 0.40 MoM*
  - *AFP > 2.0 MoM in second trimester*

- *Abnormal uterine artery Doppler (defined as mean PI >/= 2.0 in first trimester or >/= 1.45 in second trimester or bilateral notching at any gestation*

- *Unsuitable for monitoring fetal growth by clinical abdominal examination:*
  - *Maternal BMI >/= 35*
  - *Multiple fibroids or 1 large fibroid (>8cm in size)*

- *3 or more of the following minor risk factors:*
  - *Maternal age >35yrs*
  - *IVF Singleton Pregnancy*
  - *Nulliparity*
  - *BMI <20*
  - *BMI 25-34.9*
  - *Smoker 1-10/day*
  - *Previous Pre-eclampsia*
  - *Pregnancy Interval <6 months*
  - *Pregnancy Interval >60 months*

**Appendix 6: Referral Criteria for Fetal Medicine Services**

1. *For NT and first trimester anomaly scan if high risk of aneuploidy - if previous aneuploidy or maternal age > 35 years*
   *(Already above 35 years of age mentioned above)*

2. *Previous infant with structural anomaly, chromosomal abnormality, genetic syndrome or intrauterine fetal demise.*

3. *Abnormalities on index routine ultrasound*
   a. *Major or minor structural abnormality on routine ultrasound (including NT > >/= 3mm OR cystic hygroma)*
   b. *Abnormalities which may be clinically significant:*
      - *Renal pelvic dilatation ≥ 6mm before 28 weeks or ≥ 10mm after 28 weeks*
      - *Cerebral ventriculomegaly (Vp ≥ 10mm)*
      - *Enlarged cisterna magna (CM ≥ 10mm)*
      - *Echogenic bowel (as bright as bone)*
      - *Increased nuchal fold (NF ≥ 6mm)*
      - *Femur length and or head biometry < 3rd centile*
   c. *Normal variants which may be associated with risk of fetal abnormality:*
      - *Choroid plexus cysts*
      - *Echogenic foci in heart*
      - *Single umbilical artery*
   d. *Amniotic fluid abnormalities:*
      - *Oligohydramnios = SVP < 2cm or AFI of < 5cm after 26 weeks*
      - *Polyhydramnios = SVP > 8cm < 28 weeks or > 10cm > 28 weeks; or AFI > 25cm after 26 weeks*
   e. *Fetal growth pathologies:*
      - *Small babies, AC <5th centile +/- EFW < 10th centile +/- amniotic fluid Doppler abnormalities*
      - *Large babies, AC +/- EFW > 95th centile if serial scans show escalating growth*

4. *Multiple Pregnancy*
   a. *Refer if chorionicity is not clear*
   b. *All monochorionic twins, triplets, quadruplets or higher order multiple pregnancies*
   c. *Dichorionic, diamniotic twins only if added complications (meets other criteria)*

5. *Previable Prelabour Premature Rupture of Membranes (Previable PPROM) before 24 weeks gestation*

6. *Maternal / medical indications*
   a. *Diabetes – type 1 and 2 diabetes*
   b. *Epilepsy*

c. *Maternal congenital heart disease*
d. *Maternal Anti Ro or Anti La antibodies*
e. *Exposure to teratogen (if not clear discuss with Fetal Medicine physician)*
f. *Hypertension and / or chronic renal disease*
g. *Connective tissue disorders eg. SLE*
h. *Sickle cell disease*
i. *Significant chronic respiratory disease*
j. *Other medical conditions if associated with fetal pathology*

7. *Red blood cell antibodies*
   a. *Any titre if previous infant required intrauterine, top up or exchange transfusion*
   b. *All Anti-Kell*
   c. *Other antibodies if titre ≥ 32 (i.e. 32, 64, 128 etc)*

8. *Abnormal screening test result*
   a. *Combined nuchal translucency/triple test/quadruple test/non-invasive prenatal testing (NIPT)*
   b. *Toxoplasmosis, rubella, cytomegalovirus, herpes simplex (TORCH), Parvovirus B19, etc*

9. *Placental pathology – Low-lying placenta or previa with previous uterine surgery or CS / Tumour / Large haematomas / Unusual appearance (if not clear discuss with Fetal Medicine physician).*

10. *Women at with one of the following risk factors for preterm delivery*
    c. *Previous one or more spontaneous preterm delivery ≤ 34 weeks gestation (not iatrogenic)*
    d. *Previous prelabour, preterm rupture of membranes (PPROM) ≤ 34 weeks gestation*
    e. *Previous one or more mid-trimester loss 16 – 24 weeks gestation*
    f. *Known congenital uterine anomaly such as bicornuate uterus, Mullerian abnormalities etc.*

*Previous cervical surgery (previous cone biopsy or ≥ 2 large loop excision of transformation zone (LETTZ) or loop electrocautery excision procedure (LEEP)*

*Women at high risk of early onset growth restriction – previous early onset PET (</= 32 weeks); previous early onset IUGR (</= 32 weeks); previous*